

Sentiment Analysis of E-commerce Using NLP

A PROJECT REPORT

*Submitted in partial fulfillment of the requirements for
the award of the Internship - 1*

Submitted By

Nadimpalli Raja Sujith Varma

23STUCHH010477

**Under the Supervision of
Mr. SABHAPATHY M**



Department of Computer Science and Engineering (AI&ML)

ICFAI Foundation for Higher Education

Donthanapally, Shankarapalli Road, Hyderabad

India.

**FACULTY OF SCIENCE AND TECHNOLOGY IFHE
HYDERABAD**

CERTIFICATE

This is to certify that the project entitled “Sentiment Analysis of E-commerce using NLP” by Nadimpalli Raja Sujith Varma (23STUCHH010477) has been submitted in the partial fulfilment of the requirements for the award of degree of Bachelor of technology in computer Science and Engineering from Faculty of Science & Technology (FST), IFHE Hyderabad. The results embodied in this project have not been submitted to any other University or Institution for the award of any degree or diploma.

This is to certify that the above statement made by the candidates is correct to the best of my knowledge.

Internal Guide

Name:

Head of the Department

Name:

DECLARATION

I, Nadimpalli Raja Sujith Varma (Roll No: 23STUCHH010477), a student of Bachelor of Technology in Computer Science and Engineering at IFHE University, Hyderabad, hereby declare that the special project titled "Heart Disease Prediction System Using Machine Learning" submitted to the Faculty of Science and Technology is an original piece of work carried out by me under the guidance of Mr. Sabhapathy M.

I solemnly affirm that this report, in its entirety or in part, has not been previously submitted to any other institution or university for the award of any degree or diploma. Furthermore, all literature and data sources utilized in this study have been properly cited and acknowledged.

This report summarizes the work completed during the project period and represents my personal understanding and contributions.

Place: Hyderabad

Date: April 2026

Signature of the Student

Nadimpalli Raja Sujith Varma

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Nadimpalli Raja Sujith Varma

23STUCHH010477

B.Tech (CSE), IFHE University, Hyderabad

ABSTRACT

The exponential growth of e-commerce platforms has led to a substantial increase in the amount of user-generated content in terms of product-related reviews posted online by customers. It has become very common for customers to share their viewpoints on the quality, service, price, and overall satisfaction with a product. Reviews have proved to be instrumental in making purchase decisions as well as framing significant aspects of any business. Considering the sheer volume of data, it becomes extremely inefficient, time-consuming, and prone to human error to manually process these customer reviews. Natural Language Processing (NLP) has provided state-of-the-art methodologies with effective tools used for the analysis, interpretation, and processing of textual information, making it a vital phenomenon with respect to e-commerce.

The aim and objective of this project are to develop and implement an automated sentiment analysis system that can categorize e-commerce product reviews into positive, negative, and neutral classes. The data set for this project contains customer reviews collected from an e-commerce site, where the customer provides feedback in the form of ratings along with their reviews. The raw data is preprocessed using various techniques to improve data quality. The data processing techniques used for data preprocessing are transforming the data into lower case, removing special characters, removing stop words, and stemming, i.e., transforming the word into its root word.

After the preprocessing of the data, the processed data is given in the form of text. For this reason, the data needs to be converted to a numerical form using feature extraction methods. A good feature extraction mechanism for the given problem is Term Frequency-Inverse Document Frequency (TFIDF) in order to identify the importance of the terms of the reviews. However, the data is eventually given in the form of a feature using a supervised learning model. The supervised learning model is used to make predictions on new data.

Finally, the developed sentiment analysis model will be deployed through a user-friendly web application built using Streamlit. The primary objective of this system is to provide meaningful insights into customer behavior and opinions based on product reviews and feedback, helping businesses enhance customer satisfaction. Such systems enable organizations to better understand their products and services, leading to informed improvements. Overall, this project demonstrates that NLP and machine learning techniques can effectively address real-world challenges in the e-commerce domain.

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1. INTRODUCTION

1.1 Background

In recent years, the rapid growth of e-commerce platforms such as Amazon, Flipkart, and other online marketplaces has led to a massive increase in user-generated content, particularly in the form of customer reviews. These reviews play a crucial role in influencing purchasing decisions, shaping brand reputation, and providing feedback to sellers. However, the sheer volume of textual data generated daily makes it extremely difficult to manually analyze and extract meaningful insights.

To address this challenge, Natural Language Processing (NLP) has emerged as a powerful solution for understanding and analyzing human language. Sentiment analysis, a key application of NLP, focuses on identifying the emotional tone behind textual data. It enables automatic classification of text into categories such as positive, negative, or neutral sentiments.

Traditional methods of sentiment analysis relied on rule-based approaches or simple machine learning algorithms, which often struggled with context, sarcasm, and language variations. With advancements in NLP techniques, including preprocessing methods like tokenization, stopwords removal, and lemmatization, along with feature extraction techniques such as TF-IDF, more accurate and scalable systems can now be developed.

In this project, a sentiment analysis system is designed to analyze e-commerce product reviews using NLP techniques and machine learning models. The system aims to automate the process of understanding customer opinions and provide quick, reliable sentiment predictions through an interactive web application built using Streamlit.

1.2 Problem Statement

To address this challenge, Natural Language Processing (NLP) has emerged as a powerful solution for understanding and analyzing human language. Sentiment analysis, a key application of NLP, focuses on identifying the emotional tone behind textual data. It enables automatic classification of text into categories such as positive, negative, or neutral sentiments. Therefore, there is a need for an automated, efficient, and scalable system that can accurately classify customer reviews into sentiment categories.

1.3 Objectives

- To build a machine learning model that classifies e-commerce product reviews into Positive, Neutral, and Negative sentiments using NLP techniques.
- To preprocess textual data and extract meaningful features using TF-IDF vectorization with n-gram support.
- To improve model performance through hyperparameter tuning such as max features, n-gram range, regularization, and train-test split.
- To develop an interactive Streamlit web application for real-time sentiment prediction and visualization of results.

1.4 Scope of the Project

The scope of this project focuses on performing sentiment analysis on e-commerce product reviews using Natural Language Processing (NLP) techniques and machine learning models. The system is designed to analyze textual data and classify customer reviews into positive, negative, or neutral sentiments. The project includes key stages such as data collection, text preprocessing, feature extraction using techniques like TF-IDF, and model development for classification. Additionally, the project involves evaluating the performance of the model and deploying it as an interactive web application using Streamlit for real-time predictions. However, the scope is limited to text-based analysis and does not cover multimedia data such as images or videos, nor does it include advanced deep learning techniques or real-time integration with live e-commerce platforms.

1.5 Organisation of the Report

The report is structured into several chapters to present the project in a systematic and organized manner. Chapter 1 provides an introduction to the project, including background, problem statement, objectives, and scope. Chapter 2 presents the literature survey, discussing previous works and identifying research gaps related to sentiment analysis using NLP. Chapter 3 describes the proposed system and methodology, including dataset details, system architecture, and overall approach. Chapter 4 explains the implementation process, covering data preprocessing, feature extraction, model development, and the Streamlit web application. Chapter 5 presents the results and discussion, including model performance and analysis. Finally, Chapter 6 concludes the project, and Chapter 7 outlines future scope and possible improvements.

2. LITERATURE REVIEW

2.1 Introduction

Sentiment analysis has become an important area of research in the field of Natural Language Processing (NLP), especially with the rapid growth of e-commerce platforms. It focuses on identifying and classifying opinions expressed in textual data, such as customer reviews, into different sentiment categories. Many researchers have explored various approaches for sentiment analysis, ranging from traditional lexicon-based methods to advanced machine learning techniques. Earlier methods were simple but often failed to capture context and complex language patterns.

With the advancement of NLP, machine learning models combined with techniques like lemmatization, tokenization and TF-IDF have improved the accuracy of sentiment classification. This section reviews existing studies related to sentiment analysis on e-commerce data, highlighting their methodologies, key findings, and limitations, which form the basis for the proposed system.

2.2 Criteria for Selecting Literature

The research papers reviewed in this section were selected based on the following criteria:

- Relevance to sentiment analysis of textual data, especially in e-commerce or product reviews.
- Publication between recent years (2019–2024) to ensure up-to-date methodologies and trends in Natural Language Processing.
- Use of machine learning techniques such as TF-IDF, Logistic Regression, or other text classification models.
- Focus on preprocessing techniques, feature extraction methods (e.g., n-grams), and model optimization strategies.
- Inclusion of performance evaluation metrics such as accuracy, precision, recall, F1-score, and analysis of limitations.

2.3 Previous Works and Studies

[1] Bo Pang and Lillian Lee proposed a sentiment analysis system using machine learning techniques such as Naive Bayes, Maximum Entropy, and Support Vector Machines. Their study used movie review datasets to classify text into positive and negative sentiments. The results showed that machine learning approaches significantly improved classification accuracy compared to traditional methods. However, the system had limitations in handling complex linguistic structures and context.

[2] Bing Liu developed a comprehensive approach for opinion mining and sentiment analysis using Natural Language Processing techniques. The study focused on analyzing customer reviews from e-commerce platforms and extracting sentiment information. The system effectively identified opinions and sentiments but faced challenges in detecting sarcasm and implicit meanings in text.

[3] Alec Go, Richa Bhayani, and Lei Huang designed a sentiment classification system using Twitter data. They applied preprocessing techniques along with machine learning algorithms such as Naive Bayes and Maximum Entropy. The study achieved good accuracy in sentiment classification, demonstrating the effectiveness of large-scale textual data.

[4] Maas et al. proposed a deep learning-based sentiment analysis model using large-scale datasets and word embeddings. The study showed that deep learning approaches outperform

traditional machine learning models in capturing context and semantic meaning. However, the model required high computational resources and longer training time.

2.4 Research Gaps Identified

- Traditional machine learning models often fail to understand the context and deeper meaning of text. They mainly rely on surface-level features, which leads to inaccurate sentiment classification in complex sentences. This limits their effectiveness in real-world applications.
- Many existing systems do not properly handle challenges such as sarcasm, ambiguous expressions, and variations in language. As a result, the predicted sentiment may not truly reflect the actual opinion of the user.
- The performance of sentiment analysis models highly depends on the quality and size of the dataset used for training. Small or unbalanced datasets can lead to poor generalization and reduced accuracy.
- Most studies focus only on model accuracy and ignore user-friendly implementation. There is limited work on developing interactive applications that provide real-time sentiment analysis for practical use.

2.5 Summary

The reviewed studies show that sentiment analysis plays an important role in understanding customer opinions from textual data. Various machine learning and NLP techniques such as Naive Bayes, Support Vector Machines, and deep learning models have been used to improve classification accuracy. While these methods provide effective results, challenges such as handling context, sarcasm, and data quality still remain. The literature also highlights the importance of proper preprocessing and feature extraction techniques for better performance. Based on these observations, the current project focuses on using NLP techniques and machine learning models to develop an efficient and user-friendly sentiment analysis system for e-commerce reviews.

3. Methodology

3.1 Overview

The goal of this project is to build a machine learning system that can analyze customer reviews from e-commerce platforms and classify them into positive, negative, or neutral sentiments. The system takes textual data in the form of user reviews as input, processes it using Natural Language Processing (NLP) techniques, and passes it through a trained classification model to generate sentiment predictions. The output is displayed as a sentiment label along with a confidence score indicating the strength of the prediction.

Customer reviews play a significant role in influencing purchasing decisions and understanding product quality. However, analyzing a large volume of reviews manually is time-consuming and inefficient. This system helps automate the process by identifying patterns and emotions expressed in text data. The project does not aim to replace human judgment but to assist businesses and users in quickly understanding customer opinions through an easy-to-use web-based application.

3.2 Dataset Description

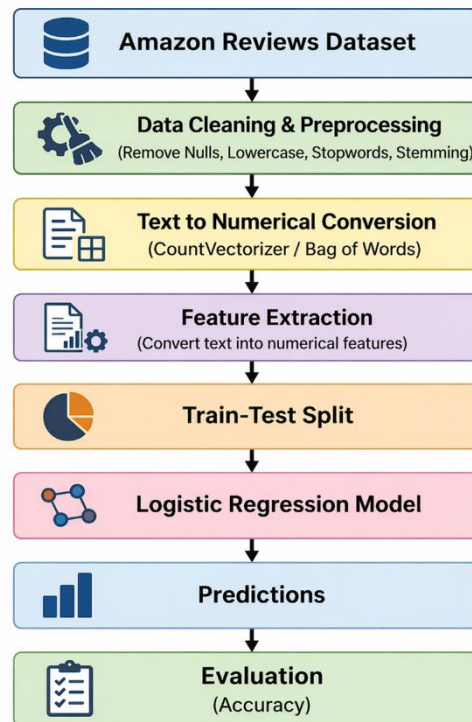
Feature Name	Type	Description
reviewId	String	Unique identifier for each review
userName	String	Name of the user (may contain missing values)
content	Text	Actual customer review text
score	Numeric	Rating given by the user (1 to 5)
thumbsUpCount	Numeric	Number of users who liked the review
reviewCreatedVersion	Categorical	Version of the app when review was created
at	DateTime	Date and time when the review was posted
appVersion	Categorical	Version of the application

The dataset includes a rating column ranging from 1 to 5, which represents customer satisfaction levels. For this project, the ratings are converted into sentiment classes: 4 and 5 as positive, 3 as neutral, and 1 and 2 as negative, to perform sentiment classification.

3.3 System Architecture

The overall system follows a structured machine learning pipeline from raw text data input to a deployed web application. The pipeline covers data collection, text preprocessing, feature extraction, model training, evaluation, and deployment. Figure 1 below summarises this workflow, and Figure 2 shows the block diagram of the system architecture.

Figure 2: Block Diagram of System Architecture



3.4 Hyperparameter Tuning

The performance of the Logistic Regression model was improved by tuning key hyperparameters through the Streamlit interface:

- Max TF-IDF Features: Controls vocabulary size (e.g., 30000)
- n-gram Range: Used (1,2,3) to capture phrases
- Regularization Parameter (C): Controls model complexity
- Test Split Ratio: Adjusted for better evaluation (e.g., 0.5)
- Random Seed: Ensures reproducibility

Tuning these parameters significantly improved model performance.

3.5 Justification for Approach

The approach used in this project is based on Natural Language Processing (NLP) and machine learning techniques, as they are well-suited for handling and analyzing textual data such as customer reviews. Text preprocessing steps like tokenization, stopword removal, and lemmatization are applied to clean and normalize the data, which helps in improving the quality of input for the model.

Machine learning mode such as Logistic Regression is chosen due to their efficiency and ability to perform well on text classification tasks. The model performance is evaluated using metrics such as accuracy to ensure reliable results.

The final system is deployed using Streamlit to provide a user-friendly interface for real-time sentiment prediction. This approach is selected because it combines simplicity, efficiency, and practical usability, making it suitable for analyzing large volumes of e-commerce reviews and providing quick insights.

4. IMPLEMENTATION

4.1 Libraries and Tools Used

The project was implemented in Python using Google Colab and Visual Studio Code. The following libraries and tools were used:

- Pandas and NumPy — for data loading, manipulation, and numerical operations.
- NLTK — for Natural Language Processing tasks such as tokenization, stopwords removal, and lemmatization.
- Scikit-learn — for TF-IDF feature extraction, machine learning models, and evaluation metrics
- Matplotlib — for visualizing data distributions and model results.

4.2 Data Pre-processing

Before training the model, the dataset was carefully cleaned and prepared to ensure better performance and accuracy. Table 2 below shows a small sample of the raw dataset to give an idea of what the data looks like before processing.

Table 2: Sample Rows from the Raw Dataset

reviewerName	reviewText	overall	summary
John	Very good product, I liked it	5	Excellent
Smith	Not worth the money	2	Bad
David	Average quality, okay to use	3	Average
Alex	Great performance and fast delivery	4	Good
Emma	Poor build quality and disappointed	1	Worst

4.3 Exploratory Data Analysis (EDA)

Before preprocessing, a visual and statistical analysis was conducted to better understand the dataset. Some key observations from the EDA are listed below:

- The dataset contains a large number of customer reviews with varying rating distributions, where positive reviews (ratings 4 and 5) are more frequent compared to negative reviews.
- The length of review text varies significantly, with longer reviews generally providing more detailed opinions and clearer sentiment.
- Reviews with higher ratings tend to contain more positive words such as “good”, “excellent”, and “best”, while lower ratings include negative words like “bad”, “poor”, and “worst”.

- Neutral reviews (rating 3) often contain mixed opinions, making them harder to classify compared to clearly positive or negative reviews.

4.4 Feature Extraction

Since machine learning algorithms cannot directly process raw text, the review text was transformed using the TF-IDF (Term Frequency–Inverse Document Frequency) technique. This method assigns weights to words based on their importance in a document relative to the entire dataset, helping the model focus on meaningful terms while reducing the impact of commonly occurring words. TF-IDF with n-gram support and max_features tuning

4.5 Train-Test Split and Scaling

After feature extraction using TF-IDF, the dataset was divided into training and testing sets to evaluate the performance of the model. 70% of the training data and the remaining 30% was used for testing. This ensures that the model is trained on a sufficient amount of data while also being evaluated on unseen data to measure its generalization ability.

Since TF-IDF already produces normalized numerical features, additional scaling is generally not required for most machine learning models used in this project. However, the feature vectors were handled carefully to maintain consistency between training and testing data. The TF-IDF vectorizer was fitted only on the training data and then applied to the test data to avoid data leakage.

4.6 Model Training

After preprocessing and feature extraction using CountVectorizer, the dataset was converted into numerical form with 30000 features. The data was split into training and testing sets, and a Decision Tree classifier was trained on the training data. The model learned patterns from the data and was then used to predict sentiments on unseen test data.

Small C → more regularization

Large C → less regularization

4.7 Streamlit Web Application

The trained sentiment analysis model was deployed using Streamlit to create an interactive web-based application. This application allows users to enter product reviews and instantly receive sentiment predictions as positive, negative, or neutral.

The user interface is designed to be simple and user-friendly, where users can input text through a text box. Once the review is submitted, the input text is preprocessed and converted into numerical features using the same Count Vectorizer used during training.

The output is displayed clearly on the screen, providing users with quick insights into the sentiment of the entered text. This web application makes the model easily accessible and demonstrates its practical use in analyzing customer feedback in real-time.

5. RESULTS

5.1 Model Comparison

The Logistic Regression used in this project was evaluated using standard performance metrics such as Accuracy. These metrics help in understanding how well the model classifies customer reviews into positive, negative, and neutral sentiments. 70% of the training data and the remaining 30% was used for testing. The model performs well on positive and negative classes but struggles with neutral class due to overlapping sentiment expressions.

Table 3: below shows the performance of the Logistic Regression model on the test dataset.

Model	Accuracy
Logistic Regression	0.70

5.2 Best Model Analysis

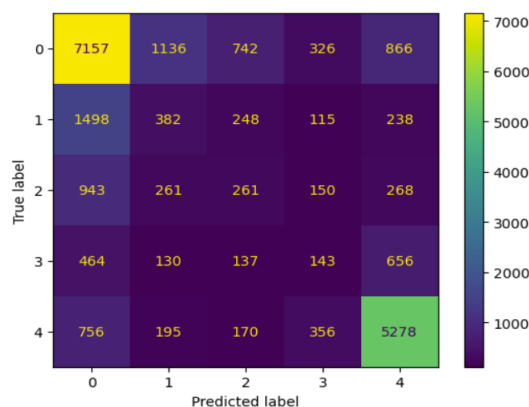
Logistic Regression was selected as the final and best-performing model for sentiment classification. The model achieved the highest testing accuracy of approximately 69.84% and demonstrated better generalization compared to other models used in this study.

Logistic Regression works by estimating the probability of a given input belonging to a particular class using a linear decision boundary. It is particularly effective for high-dimensional data such as TF-IDF features used in text classification. Its simplicity, efficiency, and ability to generalize well make it a suitable choice for sentiment analysis tasks.

5.3 Confusion Matrix

To evaluate the performance of the Decision Tree model, a confusion matrix was used. The confusion matrix provides a detailed breakdown of correct and incorrect predictions made by the model for each sentiment class (positive, negative, and neutral).

It shows how many instances were correctly classified and how many were misclassified, helping in understanding the strengths and weaknesses of the model. A higher number of values along the diagonal of the matrix indicates better model performance, as it represents correct predictions.



5.4 Feature Importance

Feature importance was analysed using the coefficients of the Logistic Regression model to understand which words contributed most to sentiment classification. Since the model was trained using TF-IDF vectorization.

The Logistic Regression model assigns coefficients to each feature, where higher positive values indicate words contributing to positive sentiment, and negative values indicate words contributing to negative sentiment. Words such as “good”, “excellent”, and “best” were associated with positive sentiment, while words like “bad”, “poor”, and “worst” contributed to negative sentiment classification.



(a) Positive



(b) Negative



(c) Neutral

5.5 Application Output

The final output of the system is generated through a Streamlit-based web application, where users can input a product review and obtain the predicted sentiment. The application processes the input text using the same preprocessing and feature extraction techniques applied during model training.

Once the user enters a review, the system analyses the text and displays the predicted sentiment as positive, negative, or neutral. The output is presented clearly on the interface, making it easy for users to understand the result. The application provides real-time predictions, demonstrating the practical usability of the model in analyzing customer feedback.

6. CONCLUSION

This project successfully developed a sentiment analysis system for e-commerce reviews using Natural Language Processing (NLP) and machine learning techniques. With the increasing amount of user-generated content on online platforms, analyzing customer opinions has become essential for both businesses and consumers. The system addresses this need by automating the process of classifying reviews into positive, negative, and neutral sentiments.

The project involved several important steps, including data collection, preprocessing, feature extraction, model training, and evaluation. Text preprocessing techniques such as tokenization, stopwords removal, and lemmatization were applied to clean and normalize the data. The CountVectorizer method was used to convert textual data into numerical form, enabling the machine learning model to process and learn from the data effectively.

A Decision Tree classifier was implemented as the core model for sentiment classification. The model was trained on the processed dataset and tested on unseen data to evaluate its performance. The results showed that the model was able to identify patterns in customer reviews and classify sentiments with satisfactory accuracy.

To enhance usability, the trained model was deployed using a Streamlit web application. This allowed users to input their own reviews and receive real-time sentiment predictions through a simple and interactive interface. The application demonstrates the practical implementation of machine learning in solving real-world problems.

Overall, this project highlights the effectiveness of combining NLP techniques with machine learning for analyzing textual data. It provides a useful tool for understanding customer feedback and supports decision-making in e-commerce environments. The system can be further improved with advanced models and larger datasets, but it successfully meets the objectives of the project.

7. FUTURE SCOPE

There are several ways this project can be extended and improved in the future:

- The current system can be enhanced by integrating advanced transformer-based models such as BERT to capture deeper contextual meaning and improve classification accuracy, especially for complex and ambiguous sentences.
- The model can be extended to perform aspect-based sentiment analysis, where sentiments are extracted for specific product features (e.g., battery, design, performance) rather than providing only an overall sentiment.
- The system can be improved to handle sarcasm, irony, and implicit sentiments, which are common challenges in natural language understanding and often lead to misclassification.
- A larger and more diverse dataset can be used to improve generalization and reduce bias, particularly for the neutral class, which is currently harder to classify accurately.
- The application can be upgraded to support multilingual sentiment analysis, allowing users to analyze reviews written in different languages using translation models or multilingual NLP techniques.
- Real-time data integration can be implemented using APIs from e-commerce platforms to continuously fetch and analyze live customer reviews.
- The web application can be enhanced with user authentication, history tracking, and dashboard analytics to provide a more personalized and professional user experience.
- Advanced visualization techniques such as interactive dashboards, trend analysis, and sentiment timelines can be added to provide deeper insights into customer feedback over time.
- The system can be optimized for deployment using scalable cloud infrastructure (e.g., Docker, AWS, or Kubernetes) to handle high traffic and ensure reliability in real-world applications.
- Future work can also include integrating explainable AI techniques (such as SHAP or LIME) to provide transparency by showing how specific words influence the sentiment prediction.

8. PROJECT TIMELINE

The project was carried out over a period of approximately 12 weeks, from January 13, 2026 to April 5, 2026. The work was divided into eight phases as shown in Table 6 below.

Table 4: Project Timeline

Phase	Duration	Task	Start Date	End Date
Phase 1	1 week	Problem understanding and literature review	Jan 13, 2026	Jan 19, 2026
Phase 2	1 week	Dataset collection and study	Jan 20, 2026	Jan 26, 2026
Phase 3	2 weeks	Data pre-processing and feature selection	Jan 27, 2026	Feb 9, 2026
Phase 4	2 weeks	Model development and training	Feb 10, 2026	Feb 23, 2026
Phase 5	1.5 weeks	Testing and accuracy comparison	Feb 24, 2026	Mar 5, 2026
Phase 6	1 week	Result analysis and visualisation	Mar 6, 2026	Mar 12, 2026
Phase 7	2 weeks	Report writing and documentation	Mar 13, 2026	Mar 26, 2026
Phase 8	10 days	Final review and submission preparation	Mar 27, 2026	Apr 5, 2026

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